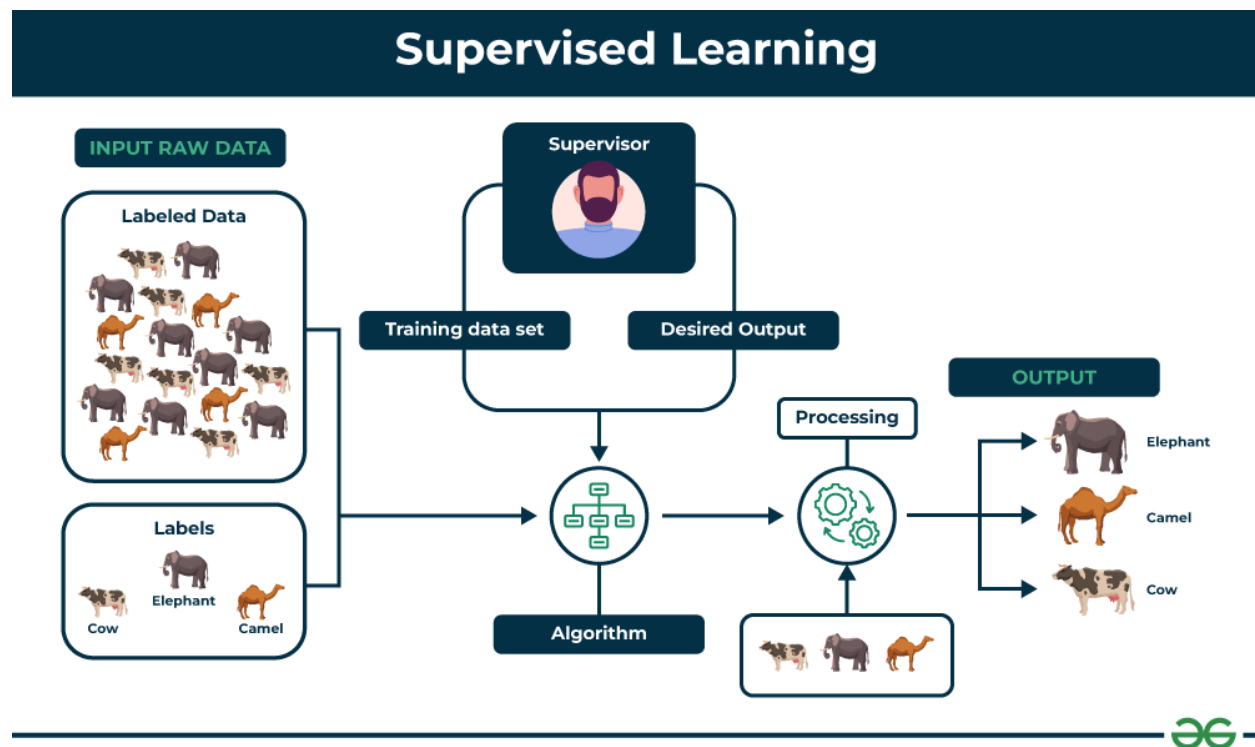


◇ Part A: Conceptual Understanding (Theory)

◇ 1. Supervised Learning Algorithms

Definition: Supervised learning is a technique where models learn from labeled data ($X \rightarrow Y$). The algorithm finds patterns between inputs and outputs to make predictions on unseen data.

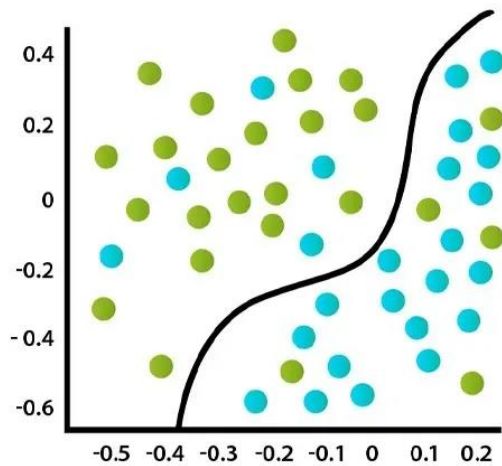
Real-life Example: Predicting house prices using features like area, bedrooms, and location score.



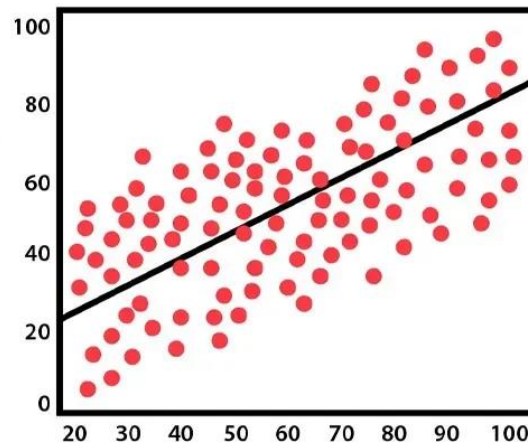
◇ 2. Regression vs Classification

- **Regression:** Predicts continuous values (e.g., house price).
- **Classification:** Predicts discrete categories (e.g., spam vs not spam).

Real-life Example: Regression → predicting salary; Classification → predicting if a loan will default.



CLASSIFICATION



REGRESSION

◇ 3. Simple Linear Regression

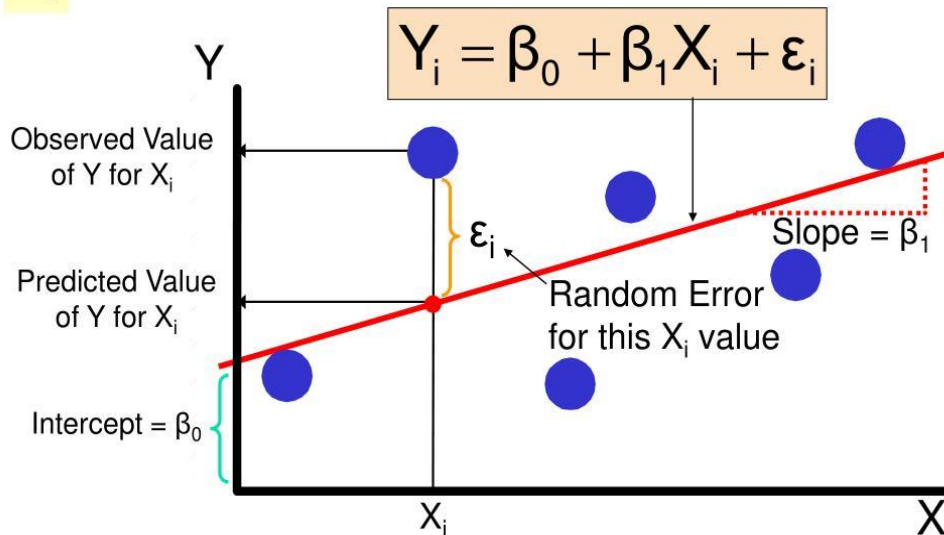
It models the relationship between one independent variable and a dependent variable using a straight line:

Fits a straight line: $y = \beta_0 + \beta_1 x$.

Real-life Example: Predicting house price based only on area.

Simple Linear Regression Model

(continued)

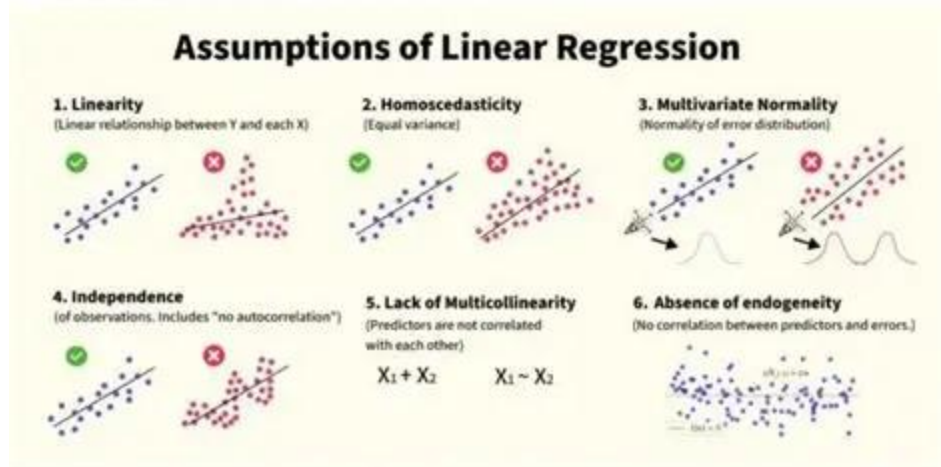


◆ 4. Assumptions of Linear Regression

Definition: Conditions required for regression results to be valid.

Explanation: Includes linearity, independence, homoscedasticity, normality of residuals, and no multicollinearity.

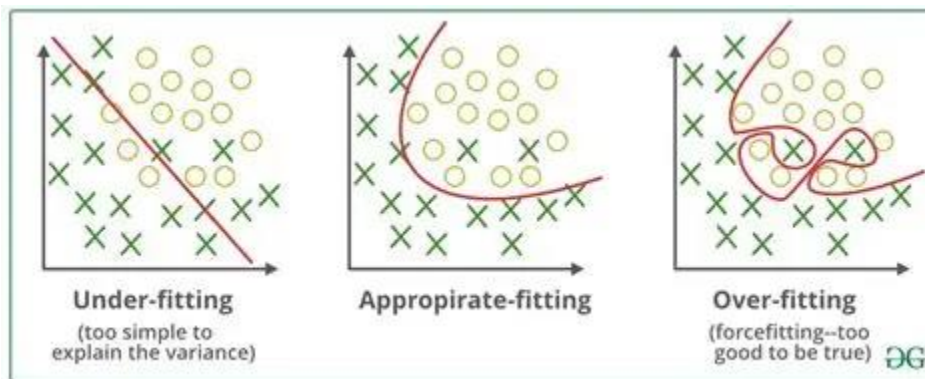
Real-life Example: If residuals are not normally distributed, predictions may be unreliable.



◇ 5. Bias–Variance Trade-Off

Bias is error due to oversimplification, variance is error due to sensitivity to training data. The trade-off balances underfitting and overfitting.

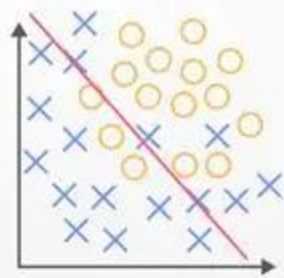
Real-life Example: A very simple model underfits, while a very complex model overfits



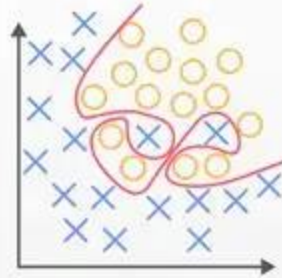
◇ 6. Overfitting vs Underfitting

- **Overfitting:** Model learns noise, performs well on training but poorly on test data.
- **Underfitting:** Model is too simple, fails to capture patterns.

Real-life Example: Overfitting → polynomial of very high degree; Underfitting → straight line for non-linear data.



Underfitting



Overfitting